Cuong Tran Dao Huy

AI ENGINEER · RESEARCHER · PhD

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Ho Chi Minh City, Vietnam

NAVIGATE

- About
- Experience
- Education
- Skills
- Courses
- Hobbies

CORE SKILLS

Research Methodology	95%
Problem Solving	93%
Computer Vision	92%
Machine Learning / Deep Learning	90%
Data Engineering	85%
Software Development (Java / C++)	60%

LANGUAGES

VN Vietnamese · GB English

~/cuong $ whoami

Cuong Tran Dao Huy

AI Researcher · Senior AI Engineer · Data Scientist

PhD in Computer Science and Engineering with expertise in **AI, Machine Learning, and Computer Vision**. Senior AI Engineer and AI content creator with various practical projects in AI and Data Science. Skilled in building and refining effective Deep Learning models - eager to drive impactful AI solutions in a professional, energetic environment.

AI RESEARCH	PHD GPA	BEST ACCURACY	ENGLISH
5+	4.3/4.5	92.7%	107
Years of experience	Sejong University	Human behavior model	TOEFL iBT score

Employment History

Senior AI Engineer

Simpson Strong-Tie

Ho Chi Minh, Vietnam

May 2024 – Present

SITUATION

Lead an AI project developing Deep Learning models for auto-assist takeoff in construction drawings (Pipeline PTO), collaborating with the FPT AI team.

PROBLEM

Manual quantitative takeoff is error-prone and time-consuming due to inconsistent scales, complex interpretations, and coordination issues.

CONTRIBUTION

Developed OCR, object detection, segmentation and line detection models to analyze 2D drawings (walls, roofs, floors) - computing both linear and area measurements.

TARGET

>90% accuracy for material estimation, significantly reducing human error and processing time.

akaOCR · PaddleOCR · YOLO · OpenCV · Azure ML · Detectron · PyTorch · Flask · FastAPI · Linux

AI Researcher & Lab Leader

Imaging & Intelligent Systems Lab - Sejong University

Seoul, South Korea

Sep 2019 – Feb 2024

5-year government-funded program - research on AI-based intelligent surveillance systems.

1 Animal Abnormal Behavior Detection 2022–2023

Problem: Detection models degrade when deployed to new surveillance environments; re-annotation is tedious.

Contribution: Iterative training framework with a binary classifier to auto-label bounding boxes - eliminating manual re-annotation.

Result: Accuracy 0.35→0.91 · mAP 0.74→0.99 in a few iterative cycles.

YOLO · OpenCV · Iterative Training · Binary Classifier · PyTorch

2 Human Abnormal Behavior Recognition 2019–2024

Problem: Low-light surveillance - reduced visibility, no color info - makes visible-only approaches ineffective.

Contribution: Built MIHAB dataset (thermal + visible). Proposed 3D multimodal feature fusion + contrastive pre-training for abnormal-behavior recognition.

Result: 92.7% accuracy (vs 85.2% baseline) · Edge-deployable on **Nvidia Jetson Orin Nano**.

3D CNNs · Contrastive Learning · Thermal Imaging · 3D Feature Fusion · Jetson Orin Nano

Editor & Co-head Content Creator

Data Science in Brief
New York, United States (Remote)

Jun 2019 - Jul 2022

Dedicated efforts to a non-profit AI / Data Science community. Oversaw editorial operations, crafted concise informative articles, organized hands-on Data Science events, and managed relationships with authors, in-house team members, and freelance contributors to foster knowledge sharing across the community.

Lecturer & Teaching Assistant

Vicohub
Ho Chi Minh, Vietnam

Oct 2018 - Sep 2019

Delivered comprehensive ML and Data Science courses at a specialized AI training center, collaborating with professors from Tan Tao University.

Linear Regression | SVM | Random Forest | PCA | K-Means | Neural Networks | NumPy | Pandas

Software Engineer

Axon Active Vietnam
Ho Chi Minh, Vietnam

Jun 2016 - Aug 2019

Built outsourcing banking applications for Swiss enterprises (Finform, Axon Fintech).

Client Onboarding (COB)
Streamlined new client intake for Postfinance, Hypothekarbank Lenzburg, Berner Kantonalbank - mobile & web.

Self-Onboarding (SOB)
Platform for bank staff to enter onboarding data per business rules - Baloise Bank SoBa, Credit Highway.

Java | Spring Framework | NGINX | Postgres | MySQL | Linux | SCRUM Agile

Education

Master & PhD - Computer Science & Engineering

Sejong University · Seoul, South Korea

Sep 2019 - Feb 2024

4.3/4.5
GPA

107
TOEFL iBT

Full Scholarship
5 Academic Years

Machine Learning | Computer Vision

Bachelor - Artificial Intelligence & Computational Science

Tan Tao University · Tay Ninh, Vietnam

Sep 2011 - Sep 2015

3.7/4.0
GPA

IELTS 6.0
TOEIC 915
English

Full Scholarship
4 Academic Years

Artificial Intelligence | Computational Science

Skills & Technologies

AI & MACHINE LEARNING

Machine Learning | Deep Learning | Computer Vision | YOLO | Object Detection | Segmentation | OCR | Contrastive Learning | 3D CNNs | Feature Fusion | PyTorch | OpenCV | Detectron | PaddleOCR

PROGRAMMING & DEV

Python | Java | C++ | Linux | Flask | FastAPI | Spring Framework | NGINX | Postgres | MySQL | NumPy | Pandas

CLOUD & INFRASTRUCTURE

- Azure Machine Learning
- Nvidia Jetson Orin Nano
- Edge AI
- Linux Server
- Windows Server

RESEARCH & METHODOLOGY

- Research Methodology
- Dataset Construction
- Iterative Training
- Model Evaluation
- Data Engineering
- Problem Solving
- SCRUM Agile

Courses & Certifications

PyImageSearch Gurus
PyImageSearch
May 2021

Data Science with Python
Vicohub
Nov 2018

Agile Leadership & Fit for Scrum
Axon Active Vietnam
May 2017

Hobbies & Interests

- Reading
- Music
- Guitar
- Sports
- Socializing
- Community